

## VIGNA DI VALLE, Italy

WMO: 162240

Lat: 42.0803N

Lon: 12.2114E

Elev: 873

StdP: 14.24

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 32.9       | 35.5 | 18.6                          | 14.5 | 38.9 | 22.3 | 17.3 | 39.9 | 25.5                  | 42.1 | 23.3 | 43.3 | 12.9                  | 30   | 0.411 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.8                   | 91.6            | 70.0 | 88.9 | 69.7 | 86.3 | 69.4 | 75.4                | 83.1 | 73.7 | 81.8 | 72.4 | 80.7 | 9.0                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.4                            | 128.1 | 79.1 | 71.4 | 119.9 | 77.4 | 69.9 | 113.4 | 76.2 | 39.5          | 82.5 | 37.9 | 81.8 | 36.6 | 80.6 | 82.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 21.5              | 18.4 | 15.8 | 30.3                       | 95.6 | 3.6                | 3.4 | 27.7                                               | 98.0 | 25.6       | 99.9 | 23.5       | 101.8 | 20.9       | 104.3 |
|                   |      |      | 26.1                       | 77.9 | 3.0                | 2.1 | 24.0                                               | 79.4 | 22.2       | 80.7 | 20.5       | 81.8  | 18.3       | 83.4  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 60.4        | 45.9  | 46.7  | 50.8  | 57.0  | 63.2  | 71.8  | 77.0  | 76.8  | 69.9  | 62.7  | 54.4  | 47.9  | (6)  |
|                                                                        | DBStd       | 12.06       | 4.62  | 5.16  | 4.87  | 4.44  | 4.99  | 5.49  | 3.92  | 4.40  | 4.34  | 4.75  | 5.14  | 4.75  | (7)  |
|                                                                        | HDD50       | 416         | 139   | 110   | 49    | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 20    | 95    | (8)  |
|                                                                        | HDD65       | 2845        | 593   | 513   | 439   | 243   | 96    | 9     | 0     | 0     | 8     | 97    | 318   | 529   | (9)  |
|                                                                        | CDD50       | 4221        | 12    | 16    | 75    | 211   | 408   | 653   | 838   | 832   | 597   | 395   | 152   | 32    | (10) |
|                                                                        | CDD65       | 1176        | 0     | 0     | 0     | 2     | 40    | 212   | 373   | 367   | 155   | 26    | 1     | 0     | (11) |
|                                                                        | CDH74       | 8397        | 0     | 0     | 0     | 11    | 192   | 1482  | 3099  | 2949  | 628   | 36    | 0     | 0     | (12) |
|                                                                        | CDH80       | 2901        | 0     | 0     | 0     | 0     | 23    | 444   | 1173  | 1141  | 119   | 1     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 6.6         | 7.0   | 7.8   | 7.4   | 6.4   | 6.2   | 6.1   | 6.2   | 6.1   | 6.3   | 6.3   | 6.7   | 7.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 39.7        | 4.7   | 4.0   | 3.2   | 3.0   | 2.2   | 1.7   | 0.7   | 2.0   | 3.7   | 4.2   | 5.7   | 5.0   | (15) |
|                                                                        | PrecMax     | 53.5        | 8.7   | 7.9   | 8.4   | 8.7   | 6.3   | 5.0   | 2.1   | 5.6   | 13.9  | 8.4   | 18.2  | 13.3  | (16) |
|                                                                        | PrecMin     | 25.8        | 0.4   | 0.2   | 0.6   | 0.9   | 0.1   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 1.2   | 1.1   | (17) |
|                                                                        | PrecStd     | 7.1         | 2.4   | 2.3   | 2.0   | 2.4   | 1.5   | 1.3   | 0.7   | 1.6   | 3.2   | 2.7   | 4.1   | 2.9   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 58.9  | 61.2  | 68.1  | 75.4  | 83.0  | 90.9  | 94.2  | 96.2  | 87.0  | 77.4  | 68.4  | 60.8  | (19) |
|                                                                        |             | MCWB        | 53.2  | 51.2  | 52.3  | 60.5  | 67.1  | 70.2  | 71.2  | 70.0  | 68.3  | 63.6  | 62.0  | 56.7  | (20) |
|                                                                        | 2%          | DB          | 56.1  | 57.8  | 63.8  | 71.6  | 79.0  | 87.9  | 91.4  | 92.6  | 83.3  | 74.4  | 65.6  | 58.0  | (21) |
|                                                                        |             | MCWB        | 52.2  | 50.6  | 53.7  | 59.3  | 64.7  | 69.4  | 69.7  | 70.3  | 68.7  | 64.4  | 60.6  | 54.1  | (22) |
|                                                                        | 5%          | DB          | 54.3  | 55.5  | 60.8  | 68.3  | 75.7  | 85.2  | 88.8  | 89.3  | 80.5  | 71.8  | 63.3  | 56.5  | (23) |
|                                                                        |             | MCWB        | 50.8  | 49.9  | 53.4  | 58.0  | 63.5  | 69.2  | 69.9  | 69.6  | 67.5  | 63.8  | 59.2  | 53.0  | (24) |
|                                                                        | 10%         | DB          | 52.5  | 53.6  | 58.3  | 65.2  | 72.6  | 82.3  | 86.3  | 86.3  | 77.4  | 69.6  | 61.2  | 54.8  | (25) |
|                                                                        |             | MCWB        | 49.2  | 49.8  | 51.9  | 56.6  | 62.4  | 68.0  | 69.6  | 69.2  | 66.3  | 63.0  | 57.7  | 51.6  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 55.3  | 54.7  | 58.2  | 63.8  | 69.9  | 74.9  | 78.1  | 77.3  | 75.0  | 68.5  | 64.7  | 58.4  | (27) |
|                                                                        |             | MCDB        | 56.8  | 57.8  | 62.0  | 71.0  | 78.1  | 83.2  | 85.1  | 83.6  | 79.2  | 71.7  | 67.1  | 59.6  | (28) |
|                                                                        | 2%          | WB          | 53.3  | 53.0  | 56.1  | 61.2  | 67.3  | 72.8  | 76.0  | 75.0  | 71.9  | 66.9  | 61.9  | 55.9  | (29) |
|                                                                        |             | MCDB        | 55.2  | 55.1  | 60.3  | 68.7  | 75.3  | 81.8  | 83.7  | 83.1  | 77.9  | 70.6  | 64.2  | 57.2  | (30) |
|                                                                        | 5%          | WB          | 51.5  | 51.6  | 54.5  | 59.2  | 65.4  | 71.2  | 74.2  | 73.4  | 70.1  | 65.5  | 60.0  | 54.1  | (31) |
|                                                                        |             | MCDB        | 53.5  | 54.3  | 58.5  | 66.2  | 73.0  | 80.6  | 82.3  | 81.9  | 76.2  | 69.5  | 62.4  | 55.6  | (32) |
|                                                                        | 10%         | WB          | 49.7  | 50.0  | 52.9  | 57.5  | 63.5  | 69.7  | 72.6  | 72.0  | 68.5  | 63.9  | 58.1  | 52.1  | (33) |
|                                                                        |             | MCDB        | 52.0  | 52.8  | 56.6  | 63.6  | 70.5  | 78.5  | 80.9  | 80.6  | 74.6  | 68.2  | 60.6  | 54.1  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.0   | 10.7  | 12.2  | 13.8  | 14.6  | 15.7  | 16.8  | 16.5  | 14.0  | 11.9  | 9.5   | 8.8   | (35) |
|                                                                        |             | MCDBR       | 10.8  | 12.9  | 16.0  | 17.6  | 18.3  | 18.7  | 19.8  | 19.9  | 16.5  | 14.2  | 10.7  | 10.0  | (36) |
|                                                                        | 5% WB       | MCWBR       | 8.5   | 9.2   | 10.1  | 10.5  | 9.7   | 8.7   | 8.4   | 8.2   | 8.2   | 8.2   | 7.7   | 7.6   | (37) |
|                                                                        |             | MCWBR       | 9.2   | 10.8  | 13.7  | 15.7  | 16.8  | 17.1  | 17.0  | 16.6  | 14.4  | 12.4  | 9.7   | 8.3   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.330 | 0.350 | 0.385 | 0.415 | 0.420 | 0.414 | 0.406 | 0.406 | 0.397 | 0.384 | 0.361 | 0.331 | (40) |
|                                                                        |             | taud        | 2.469 | 2.395 | 2.298 | 2.226 | 2.254 | 2.315 | 2.339 | 2.352 | 2.373 | 2.394 | 2.397 | 2.468 | (41) |
|                                                                        | Ebn at Noon |             | 258   | 269   | 272   | 271   | 272   | 273   | 274   | 271   | 265   | 253   | 241   | 246   | (42) |
|                                                                        |             | Edh at Noon | 24    | 30    | 37    | 43    | 43    | 40    | 39    | 37    | 34    | 30    | 26    | 22    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 554   | 843   | 1230  | 1638  | 2004  | 2280  | 2341  | 2043  | 1497  | 1015  | 622   | 496   | (44) |
|                                                                        | RadStd      |             | 55    | 84    | 142   | 119   | 170   | 100   | 108   | 108   | 74    | 88    | 58    | 54    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (16 neighbors) | +0.76 | N/A     | N/A    | +1.06   | N/A   | N/A   | N/A         | -158  | +252  | +124  |

Nomenclature: See separate page